



JKBOSE 2026

CLASS 11TH

STRAIGHT LINES

MATHS

LECTURE-8

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TOPICS TO BE COVERED

1

Miscellaneous Exercise

2

Related Problems

**MISCELLANEOUS
EXERCISE ON
CHAPTER – 9**

QUESTION

$$\longleftrightarrow y=c \quad \text{or} \quad \updownarrow x=c$$

Find the values of k for which the line $(k-3)x - (4-k^2)y + k^2 - 7k + 6 = 0$ is.

- (a) Parallel to the x -axis, ($y=c$)
- (b) Parallel to the y -axis,
- (c) Passing through the origin.

(Sol) Given line is:

$$(k-3)x - (4-k^2)y + (k^2 - 7k + 6) = 0 \quad \text{--- (1)}$$

(i) If (1) is parallel to x -axis, then

$$k-3=0$$

$$\Rightarrow k=3$$

(ii) If (1) is parallel to y -axis, then

$$-(4-k^2) = 0$$

$$\Rightarrow 4-k^2=0$$

$$\Rightarrow k^2=4$$

$$\Rightarrow k = \pm 2$$

(iii) If (1) passes through the origin (0,0), then

$$(k-3) \times 0 - (4-k^2) \times 0 + k^2 - 7k + 6 = 0$$

$$\Rightarrow k^2 - 7k + 6 = 0$$

$$\Rightarrow (k-6)(k-1) = 0$$

$$\Rightarrow k = 1, 6 \quad \text{Hence the result.}$$

QUESTION

Find the equations of the lines, which cut-off intercepts on the axes whose sum and product are 1 and -6, respectively.

(Sol) let the required line be: $\frac{x}{a} + \frac{y}{b} = 1$ — (1)

According to the Question, we have

$$a+b=1 \Rightarrow b=1-a \text{ — (2)}$$

$$\& ab=-6 \text{ — (3)}$$

Using (2) in (3), we get:

$$a(1-a) = -6$$

$$\Rightarrow a - a^2 = -6$$

$$\Rightarrow a^2 - a - 6 = 0$$

$$\Rightarrow (a-3)(a+2) = 0$$

$$\Rightarrow a = 3, -2$$

So, from (2), we get

$$b = 1-3, 1+2$$

$$\Rightarrow b = -2, 3$$

So from (1), the required equations are:

$$\frac{x}{3} + \frac{y}{-2} = 1 \text{ or } \frac{x}{-2} + \frac{y}{3} = 1$$

$$\Rightarrow \frac{-2x+3y}{6} = 1 \text{ or } \frac{3x-2y}{6} = 1$$

$$\begin{aligned} -2x+3y &= 6 \\ \text{or } 3x-2y &= 6 \end{aligned}$$

$$\Rightarrow \begin{aligned} -2x+3y-6 &= 0 \\ \text{or } 3x-2y-6 &= 0 \end{aligned}$$

Hence the result.

QUESTION

What are the points on the y -axis whose distance from the line $\frac{x}{3} + \frac{y}{4} = 1$ is 4 units.

(Homework Problem)

QUESTION

$$\cos\left(\theta - \frac{\phi + \theta}{2}\right) = \cos\left(\frac{2\theta - \phi - \theta}{2}\right) = \cos\left(\frac{\theta - \phi}{2}\right)$$



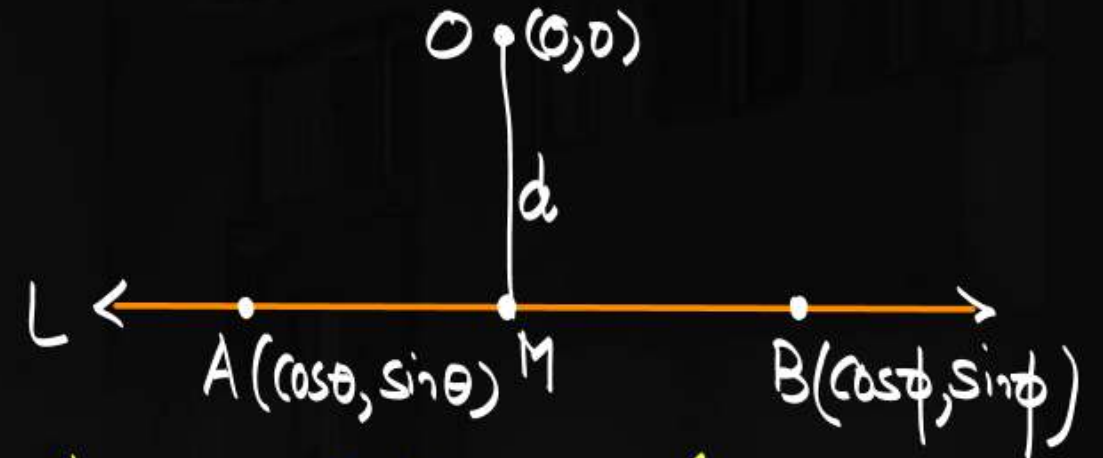
Find **perpendicular distance** from the **origin** to the line joining the points **$(\cos\theta, \sin\theta)$** and **$(\cos\phi, \sin\phi)$**

(Sol) The equation of the line joining $(\cos\theta, \sin\theta)$ & $(\cos\phi, \sin\phi)$ is;

$$y - \sin\theta = \left(\frac{\sin\phi - \sin\theta}{\cos\phi - \cos\theta}\right)(x - \cos\theta)$$

$$\Rightarrow y - \sin\theta = \frac{\cancel{2} \cos\left(\frac{\phi + \theta}{2}\right) \cancel{\sin\left(\frac{\phi - \theta}{2}\right)}}{\cancel{-2} \sin\left(\frac{\phi + \theta}{2}\right) \cancel{\sin\left(\frac{\phi - \theta}{2}\right)}} (x - \cos\theta)$$

$$\Rightarrow \sin\left(\frac{\phi + \theta}{2}\right)y - \sin\theta \sin\left(\frac{\phi + \theta}{2}\right) = -\cos\left(\frac{\phi + \theta}{2}\right)x + \cos\theta \cos\left(\frac{\phi + \theta}{2}\right) = 0$$



$$\Rightarrow \cos\left(\frac{\phi + \theta}{2}\right)x + \sin\left(\frac{\phi + \theta}{2}\right)y - \left(\cos\theta \cos\left(\frac{\phi + \theta}{2}\right) + \sin\theta \sin\left(\frac{\phi + \theta}{2}\right)\right) = 0$$

$$\Rightarrow \boxed{\cos\left(\frac{\phi + \theta}{2}\right)x + \sin\left(\frac{\phi + \theta}{2}\right)y - \cos\left(\frac{\theta - \phi}{2}\right) = 0} \quad \text{--- (1)}$$

Suppose that d is the distance (perpendicular) of $(0,0)$ from (1), then.

$$d = \frac{\left| \cos\left(\frac{\phi+\theta}{2}\right) \times 0 + \sin\left(\frac{\phi+\theta}{2}\right) \times 0 - \cos\left(\frac{\theta-\phi}{2}\right) \right|}{\sqrt{\cos^2\left(\frac{\phi+\theta}{2}\right) + \sin^2\left(\frac{\phi+\theta}{2}\right)}}$$

$$\Rightarrow d = \frac{\left| -\cos\left(\frac{\theta-\phi}{2}\right) \right|}{\sqrt{1}}$$

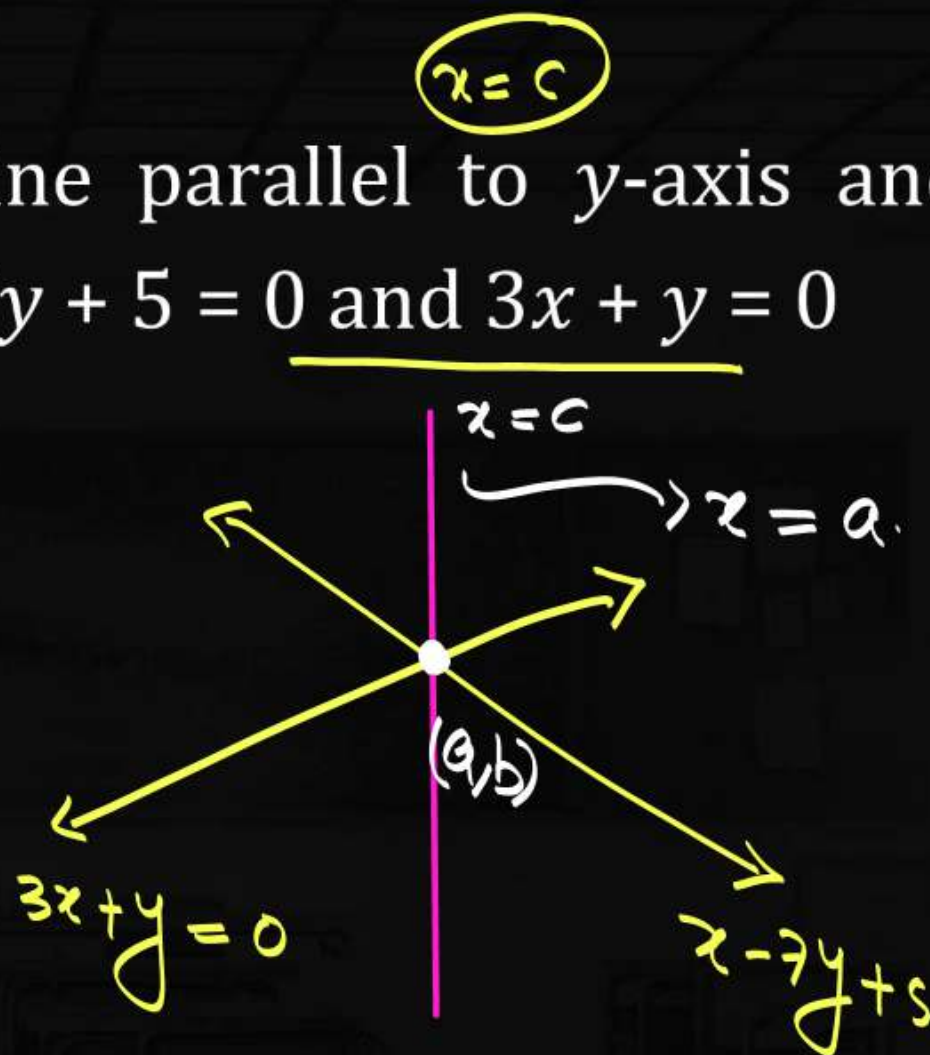
$$\Rightarrow d = \left| \cos\left(\frac{\theta-\phi}{2}\right) \right|$$

Hence the result.

QUESTION

Find the equation of the line parallel to y -axis and drawn through the point of intersection of the lines $x - 7y + 5 = 0$ and $3x + y = 0$

(Homework Problem)



QUESTION

Find the equation of a line drawn perpendicular to the line $\frac{x}{4} + \frac{y}{6} = 1$ through the point, where it meets the y -axis.

QUESTION

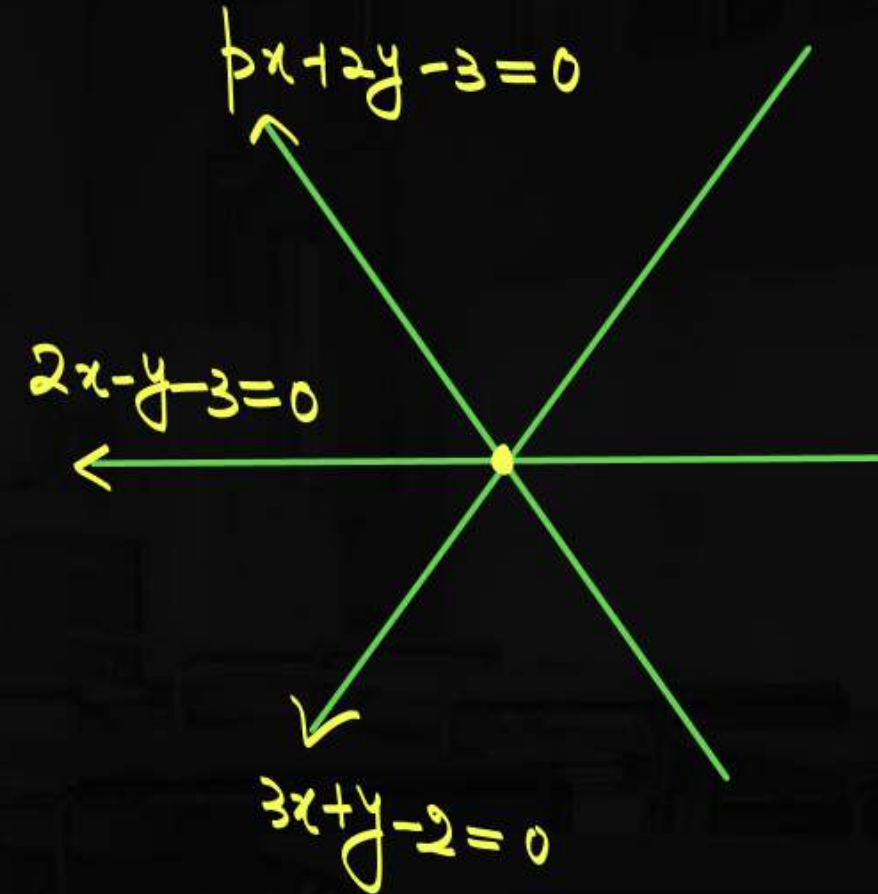
Find the area of the triangle formed by the lines $y - x = 0$, $x + y = 0$ and $x - k = 0$

QUESTION

Find the value of p so that the three lines $3x + y - 2 = 0$, $px + 2y - 3 = 0$ and $2x - y - 3 = 0$ may intersect at one point.

- (a) $p = 4$ ✗
- (b) $p = 3$ ✗
- (c) $p = 5$ ✓
- (d) $p = -6$ ✗

Homework.



QUESTION

If three lines whose equations are $y = m_1x + c_1$, $y = m_2x + c_2$ and $y = m_3x + c_3$ are concurrent, then show that $m_1(c_2 - c_3) + m_2(c_3 - c_1) + m_3(c_1 - c_2) = 0$.

Proof: Given lines are

$$y = m_1x + c_1 \text{ --- (1)}$$

$$y = m_2x + c_2 \text{ --- (2)}$$

$$y = m_3x + c_3 \text{ --- (3)}$$

Subtracting (1) & (2), we get;

$$\cancel{y} - \cancel{y} = m_1x + c_1 - m_2x - c_2$$

$$\Rightarrow (m_1 - m_2)x + (c_1 - c_2) = 0$$

$$\Rightarrow (m_1 - m_2)x = c_2 - c_1$$

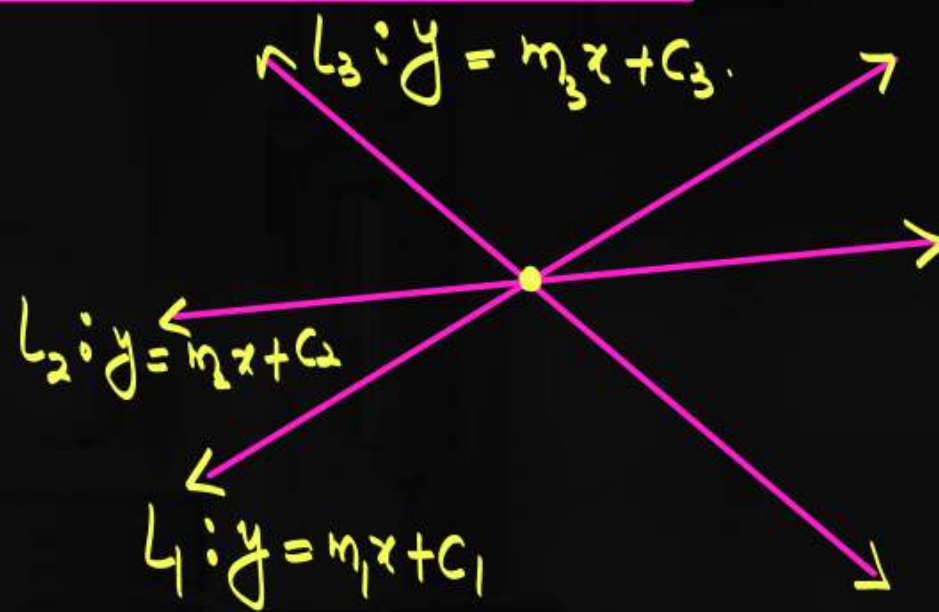
$$\Rightarrow x = \frac{c_2 - c_1}{m_1 - m_2}$$

So from (1), we get;

$$y = m_1 \left(\frac{c_2 - c_1}{m_1 - m_2} \right) + c_1$$

$$\Rightarrow y = \frac{m_1c_2 - \cancel{m_1c_1} + \cancel{c_1m_1} - c_1m_2}{m_1 - m_2}$$

$$\Rightarrow y = \frac{m_1c_2 - c_1m_2}{m_1 - m_2}$$



So $\left(\frac{c_2 - c_1}{m_1 - m_2}, \frac{m_1c_2 - c_1m_2}{m_1 - m_2} \right)$ is the point of intersection of (1) & (2).

Since ①, ② & ③ are Concurrent lines, then the point of intersection of ① & ② lies on ③. So,

$$\frac{m_1 c_2 - c_1 m_2}{m_1 - m_2} = m_3 \left(\frac{c_2 - c_1}{m_1 - m_2} \right) + c_3$$

$$\Rightarrow \frac{m_1 c_2 - c_1 m_2}{\cancel{m_1 - m_2}} = \frac{m_3 (c_2 - c_1) + c_3 (\cancel{m_1 - m_2})}{\cancel{(m_1 - m_2)}}$$

$$\Rightarrow m_1 c_2 - c_1 m_2 - m_3 (c_2 - c_1) - c_3 (m_1 - m_2) = 0$$

$$\Rightarrow m_1 c_2 - c_1 m_2 + m_3 (c_1 - c_2) - c_3 m_1 + c_3 m_2 = 0$$

$$\Rightarrow m_1 (c_2 - c_3) + m_2 (c_3 - c_1) + m_3 (c_1 - c_2) = 0$$

Hence proved.

QUESTION

Find the equation of the lines through the point $(3, 2)$ which make an angle of 45° with the line $x - 2y = 3$.

(Sol) let the slope of the required line be m . Then

$$\tan \theta = \left| \frac{m - \frac{1}{2}}{1 + \frac{1}{2}m} \right| = \left| \frac{2m - 1}{m + 2} \right|$$

$$\Rightarrow \left| \frac{2m - 1}{m + 2} \right| = \tan 45^\circ = 1.$$

$$\Rightarrow \frac{2m - 1}{m + 2} = 1, -1.$$

$$\Rightarrow 2m - 1 = m + 2, -m - 2$$

$$\Rightarrow 2m - m = 2 + 1, 2m + m = -2 + 1$$

$$\Rightarrow m = 3, -\frac{1}{3}$$

So, the required lines are:

$$y - 2 = 3(x - 3)$$

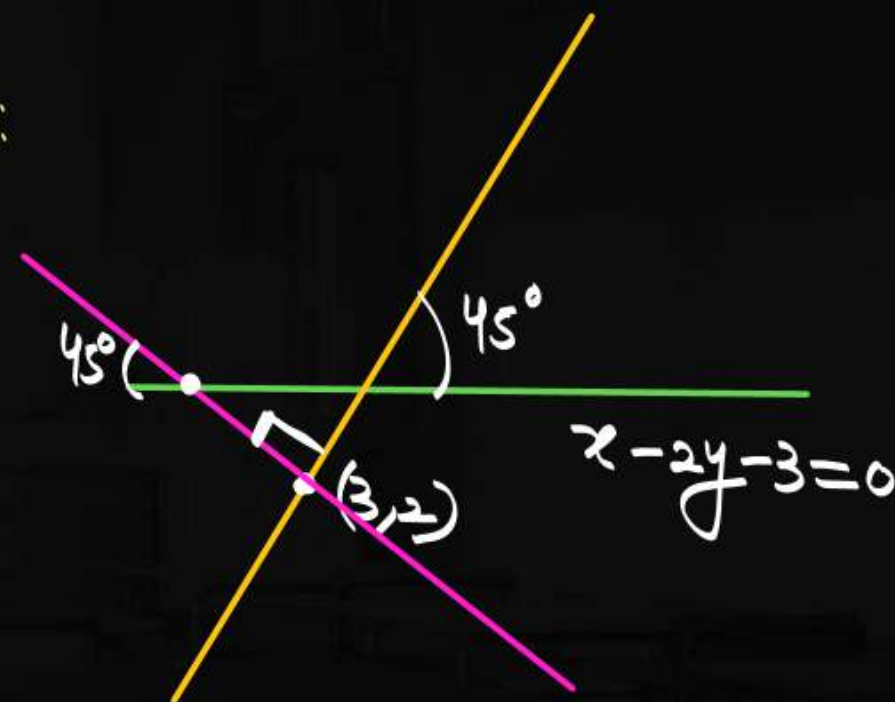
$$\text{or } y - 2 = -\frac{1}{3}(x - 3)$$

$$\Rightarrow y - 2 = 3x - 9$$

$$\text{or } 3y - 6 = -x + 3$$

$$\Rightarrow 3x - y - 7 = 0$$

$$\text{or } x + 3y - 9 = 0.$$



Hence the result.

QUESTION

Find the equation of the line passing through the point of intersection of the lines $4x + 7y - 3 = 0$ and $2x - 3y + 1 = 0$ that has equal intercepts on the axes.

(H/W).

QUESTION

Show that the equation of the line passing through the origin and making an angle θ

with the line $y = mx + c$ is $\frac{y}{x} = \frac{m \pm \tan \theta}{1 \mp m \tan \theta}$

Proof: We know that $\tan \theta = \left| \frac{m' - m}{1 + m'm} \right|$ where m' is the slope of the required line.

$$\Rightarrow \pm \tan \theta = \frac{m' - m}{1 + m'm}$$

$$\Rightarrow \pm \tan \theta + m'm \tan \theta = m' - m$$

$$\Rightarrow m \pm \tan \theta = m' \mp m' m \tan \theta$$

$$\Rightarrow m'(1 \mp m \tan \theta) = m \pm \tan \theta$$

$$y = mx$$

$$\Rightarrow m' = \frac{m \pm \tan \theta}{1 \mp m \tan \theta}$$

Therefore the required line is:

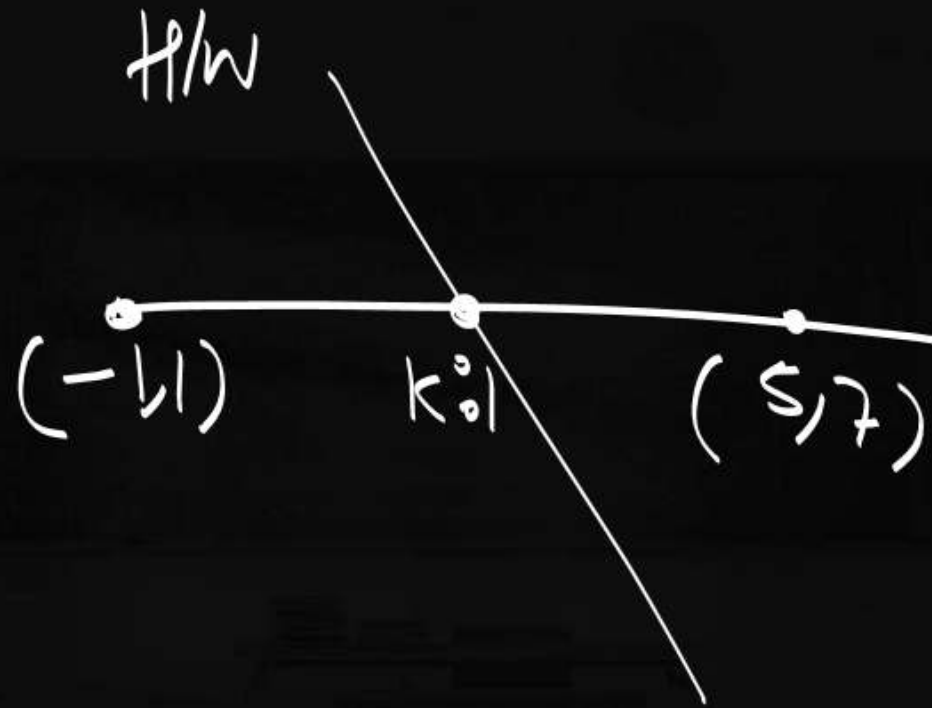
$$y = \left(\frac{m \pm \tan \theta}{1 \mp \tan \theta} \right) x$$

$$\Rightarrow \frac{y}{x} = \frac{m \pm \tan \theta}{1 \mp \tan \theta}$$

Hence proved.

QUESTION

In what ratio, the line joining $(-1, 1)$ and $(5, 7)$ is divided by the line $x + y = 4$?



QUESTION

Find the distance of the line $4x + 7y + 5 = 0$ from the point $(1, 2)$ along the line $2x - y = 0$.

H/W

QUESTION

Find the direction in which a straight line must be drawn through the point $(-1, 2)$ so that its point of intersection with the line $x + y = 4$ may be at a distance of 3 units from this point. (H/W).

QUESTION

The hypotenuse of a right angled triangle has its ends at the points $(1, 3)$ and $(-4, 1)$. Find an equation of the legs (perpendicular sides) of the triangle which are parallel to the axes.

HP/W

QUESTION

Find the image of the point $(3, 8)$ with respect to the line $x + 3y = 7$ assuming the line to be a plane mirror.

(Sol) Suppose that $Q(h, k)$ be the image of the point $P(3, 8)$ w.r.t. the line

$$x + 3y - 7 = 0 \quad \text{--- (1)}$$

Here clearly M is the mid-point of PQ ,
So by mid-point formula,

$$M(x, y) = M\left(\frac{h+3}{2}, \frac{k+8}{2}\right)$$

Since $M\left(\frac{h+3}{2}, \frac{k+8}{2}\right)$ lies on (1), so

$$\frac{h+3}{2} + 3\left(\frac{k+8}{2}\right) - 7 = 0$$

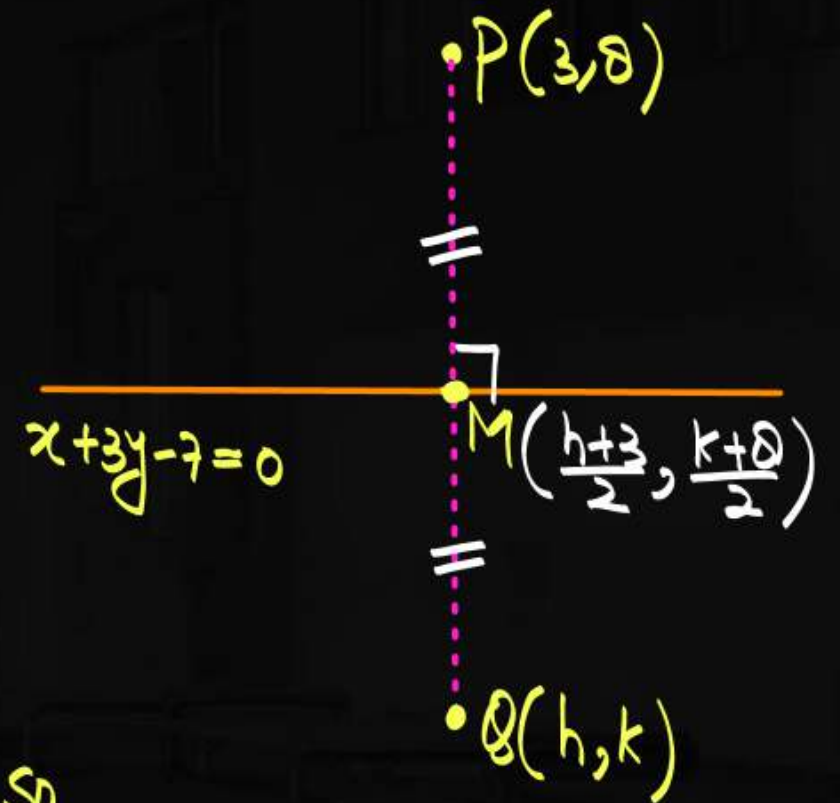
$$\Rightarrow \frac{h+3 + 3k+24}{2} - 7 = 0$$

$$\Rightarrow h + 3k + 27 - 14 = 0$$

$$\Rightarrow h + 3k + 13 = 0 \quad \text{--- (2)}$$

Also PQ is perpendicular on (1), so

$$\left(\frac{k-8}{h-3}\right) \times \frac{1}{3} = -1$$



$$\Rightarrow \frac{k-8}{h-3} = 3.$$

$$\Rightarrow k = 3(h-3) + 8$$

$$\Rightarrow k = 3h - 9 + 8$$

$$\Rightarrow \boxed{k = 3h - 1} \text{---(3)}$$

Using (3) in (2), we get;

$$h + 3(3h - 1) + 13 = 0$$

$$\Rightarrow h + 9h - 3 + 13 = 0$$

$$\Rightarrow 10h + 10 = 0$$

$$\Rightarrow \boxed{h = -1}.$$

So from (3), we get;

$$k = 3 \times -1 - 1$$

$$\Rightarrow k = -4.$$

$\Rightarrow$ the image of $P(3, 8)$ is $Q(-1, -4)$

Hence the result.

QUESTION

If the lines $y = 3x + 1$ and $2y = x + 3$ are equally inclined to the line $y = mx + 4$ find the value of m .

H/W

QUESTION

If sum of the perpendicular distances of a variable point $P(x, y)$ from the lines $x + y - 5 = 0$ and $3x - 2y + 7 = 0$ is always 10. Show that P must move on a line.

H/W

QUESTION

Find equation of the line which is equidistant from parallel lines $9x + 6y - 7 = 0$ and $3x + 2y + 6 = 0$.

H/W

QUESTION

A ray of light passing through the point $(1, 2)$ reflects on the x -axis at point A and the reflected ray passes through the point $(5, 3)$. Find the coordinates of A.

(Already done)

H/W

QUESTION

Prove that the product of the lengths of the perpendiculars drawn from the points $(\sqrt{a^2 - b^2}, 0)$ and $(-\sqrt{a^2 - b^2}, 0)$ to the line $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$ is b^2 .

H/W

QUESTION

A person standing at the junction (crossing) of two straight paths represented by the equations $2x - 3y + 4 = 0$ and $3x + 4y - 5 = 0$ wants to reach the path whose equation is $6x - 7y + 8 = 0$ in the least time. Find equation of the path that he should follow.

(IP/W).

QUESTION (NCERT EXAMPLAR):

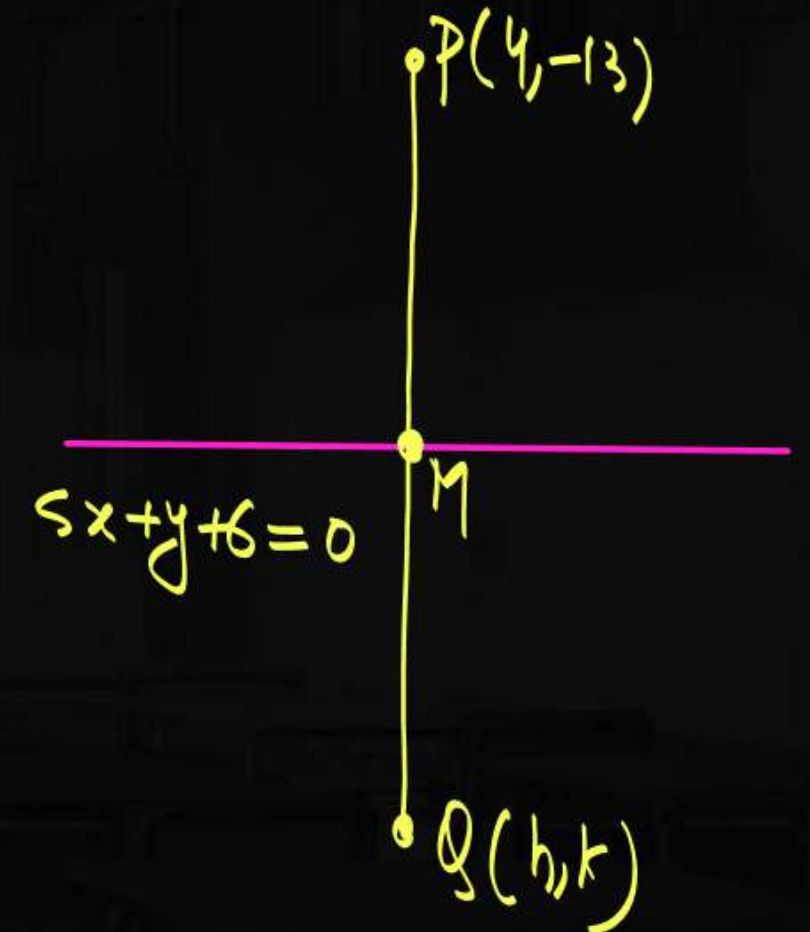
The reflection of the point $(4, -13)$ about the line $5x + y + 6 = 0$ is

1 $(-1, -14)$ ✓

2 $(3, 4)$ ✗

3 $(0, 0)$ ✗

4 $(1, 2)$ ✗



QUESTION (NCERT EXAMPLAR):

The point $(4, 1)$ undergoes the following two successive transformations:

- (i) Reflection about the line $y = x$
- (ii) Translation through a distance 2 units along the positive x -axis

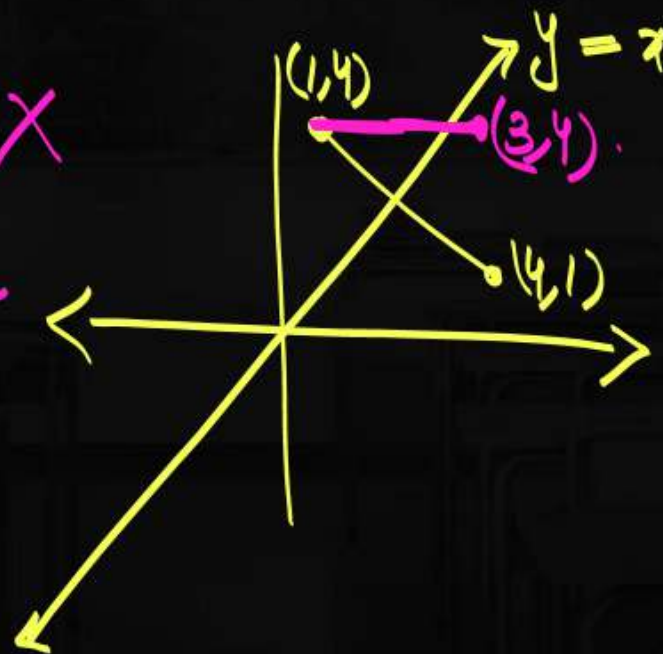
Then the final coordinates of the point are

1 $(-1, -14)$ ✗

2 $(3, 4)$ ✓

3 $(0, 0)$ ✗

4 $(1, 2)$ ✗



Firdousa Majeed firdouso 

Syed Fakhira 

Zuha Nazir

Thank
You